# Math for ML Cheat Sheet: Linear Algebra (v1.1)



## Notation

| | |
|---|---|
| $\boldsymbol{x} \in \mathbb{R}^n$ | column vector, entries $x_1, \ldots, x_n$ (Strang: $\mathbf{x}$ or $x$) |
| $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ | matrix, $m$ rows, $n$ columns, entry $a_{ij}$ (Strang: $A$) |
| $\boldsymbol{a}_j, \boldsymbol{a}_i^\mathsf{T}$ | column $j$ / row $i$ of $\boldsymbol{A}$ |
| $\boldsymbol{x}^\mathsf{T}$ | transpose: row vector |
| $\boldsymbol{e}_i$ | $i$-th standard basis vector; $\boldsymbol{I}$ = identity |
| $\boldsymbol{x}^\mathsf{T}\boldsymbol{y} = \boldsymbol{x} \cdot \boldsymbol{y} = \langle \boldsymbol{x}, \boldsymbol{y} \rangle$ | dot (inner) product |

## Vectors

**Linear combination** of $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k \in \mathbb{R}^n$ with scalars $c_1, \ldots, c_k$:

$$c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 + \cdots + c_k\boldsymbol{v}_k.$$

$\operatorname{span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\}$ = set of *all* linear combinations. The vectors are *linearly independent* if $\sum_i c_i\boldsymbol{v}_i = \boldsymbol{0}$ only when all $c_i = 0$.

**Dot product**

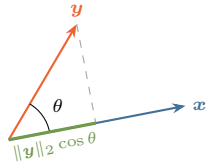
$$\boldsymbol{x}^\mathsf{T}\boldsymbol{y} = \sum_{i=1}^{n} x_i y_i = x_1 y_1 + \cdots + x_n y_n \in \mathbb{R}.$$

Symmetric, linear in each argument, $\boldsymbol{x}^\mathsf{T}\boldsymbol{x} = \|\boldsymbol{x}\|_2^2 \geq 0$.

**Dot product and angle**

$$\boldsymbol{x}^\mathsf{T}\boldsymbol{y} = \|\boldsymbol{x}\|_2 \, \|\boldsymbol{y}\|_2 \cos\theta$$

$$\cos\theta = \frac{\boldsymbol{x}^\mathsf{T}\boldsymbol{y}}{\|\boldsymbol{x}\|_2 \|\boldsymbol{y}\|_2}$$




$\boldsymbol{x}^\mathsf{T}\boldsymbol{y} > 0$: acute, $= 0$: **orthogonal** ($\boldsymbol{x} \perp \boldsymbol{y}$), $< 0$: obtuse.

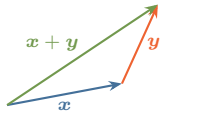
**Cauchy–Schwarz inequality**

$$|\boldsymbol{x}^\mathsf{T}\boldsymbol{y}| \leq \|\boldsymbol{x}\|_2 \, \|\boldsymbol{y}\|_2$$

with equality iff $\boldsymbol{x}, \boldsymbol{y}$ are parallel. (Equivalent to $|\cos\theta| \leq 1$.)

**Norm** A function $\|\cdot\| : \mathbb{R}^n \to \mathbb{R}$ with, for all $\boldsymbol{x}, \boldsymbol{y}, \alpha$:

1. $\|\boldsymbol{x}\| \geq 0$ and $\|\boldsymbol{x}\| = 0 \iff \boldsymbol{x} = \boldsymbol{0}$
2. $\|\alpha\boldsymbol{x}\| = |\alpha| \, \|\boldsymbol{x}\|$
3. $\|\boldsymbol{x} + \boldsymbol{y}\| \leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$ (**triangle**)

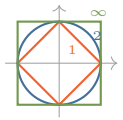



a side is never longer than the other two combined

Common norms:

$\|\boldsymbol{x}\|_2 = \sqrt{\boldsymbol{x}^\mathsf{T}\boldsymbol{x}} = \sqrt{\sum_i x_i^2}$ (Euclidean, default)
$\|\boldsymbol{x}\|_1 = \sum_i |x_i|$ (Manhattan)
$\|\boldsymbol{x}\|_\infty = \max_i |x_i|$ (max)

**Unit vector** $\boldsymbol{u} = \boldsymbol{x}/\|\boldsymbol{x}\|_2$ has $\|\boldsymbol{u}\|_2 = 1$.




unit balls $\|\boldsymbol{x}\| \leq 1$

## Matrix Multiplication

$\boldsymbol{A} \in \mathbb{R}^{m \times n}, \boldsymbol{B} \in \mathbb{R}^{n \times p} \Rightarrow \boldsymbol{C} = \boldsymbol{AB} \in \mathbb{R}^{m \times p}$. Inner dimensions must agree. In general $\boldsymbol{AB} \neq \boldsymbol{BA}$; but $(\boldsymbol{AB})\boldsymbol{C} = \boldsymbol{A}(\boldsymbol{BC})$.

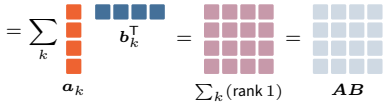
**As inner products** (row · column)

$$c_{ij} = \boldsymbol{a}_i^\mathsf{T}\boldsymbol{b}_j = \sum_{k=1}^{n} a_{ik} b_{kj}$$



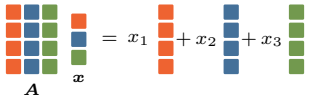

**As outer products** (column × row, sum of rank-one matrices)

$$\boldsymbol{AB} = \sum_{k=1}^{n} \boldsymbol{a}_k \, \boldsymbol{b}_k^\mathsf{T} \qquad (\boldsymbol{a}_k = \text{col } k \text{ of } \boldsymbol{A},\ \boldsymbol{b}_k^\mathsf{T} = \text{row } k \text{ of } \boldsymbol{B})$$




**Matrix–vector: combination of columns** (Strang's view)

$$\boldsymbol{Ax} = x_1\boldsymbol{a}_1 + x_2\boldsymbol{a}_2 + \cdots + x_n\boldsymbol{a}_n$$




## Transpose

$(\boldsymbol{A}^\mathsf{T})_{ij} = a_{ji}$: rows become columns.

- $(\boldsymbol{A}^\mathsf{T})^\mathsf{T} = \boldsymbol{A}$, $(\boldsymbol{A} + \boldsymbol{B})^\mathsf{T} = \boldsymbol{A}^\mathsf{T} + \boldsymbol{B}^\mathsf{T}$
- $(\boldsymbol{AB})^\mathsf{T} = \boldsymbol{B}^\mathsf{T}\boldsymbol{A}^\mathsf{T}$ (order reverses)
- $\boldsymbol{x}^\mathsf{T}\boldsymbol{y}$ is a scalar; $\boldsymbol{x}\boldsymbol{y}^\mathsf{T}$ is a matrix
- $(\boldsymbol{Ax})^\mathsf{T}\boldsymbol{y} = \boldsymbol{x}^\mathsf{T}(\boldsymbol{A}^\mathsf{T}\boldsymbol{y})$




## Special Matrices



**Symmetric** $\boldsymbol{A} = \boldsymbol{A}^\mathsf{T}$, i.e. $a_{ij} = a_{ji}$ (square). $\boldsymbol{A}^\mathsf{T}\boldsymbol{A}$ and $\boldsymbol{A}\boldsymbol{A}^\mathsf{T}$ are always symmetric. Symmetric matrices have real eigenvalues and orthogonal eigenvectors (see p. 2).

**Inverse** $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ is invertible if there is $\boldsymbol{A}^{-1}$ with $\boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{A}\boldsymbol{A}^{-1} = \boldsymbol{I}$. Then $\boldsymbol{Ax} = \boldsymbol{b}$ has the unique solution $\boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{b}$.

- invertible $\iff \det \boldsymbol{A} \neq 0 \iff$ columns independent $\iff \boldsymbol{Ax} = \boldsymbol{0}$ only for $\boldsymbol{x} = \boldsymbol{0}$
- $(\boldsymbol{AB})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$, $(\boldsymbol{A}^\mathsf{T})^{-1} = (\boldsymbol{A}^{-1})^\mathsf{T}$
- $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

## Orthogonal Matrices

**Orthogonal matrix** Square $\boldsymbol{Q}$ with *orthonormal* columns ($\boldsymbol{q}_i^\mathsf{T}\boldsymbol{q}_j = 0$ for $i \neq j$, $\|\boldsymbol{q}_i\|_2 = 1$):

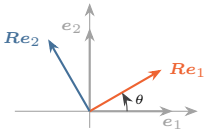
$$\boldsymbol{Q}^\mathsf{T}\boldsymbol{Q} = \boldsymbol{Q}\boldsymbol{Q}^\mathsf{T} = \boldsymbol{I}, \qquad \boldsymbol{Q}^{-1} = \boldsymbol{Q}^\mathsf{T}.$$




Preserves lengths and angles: $\|\boldsymbol{Qx}\|_2 = \|\boldsymbol{x}\|_2$, $(\boldsymbol{Qx})^\mathsf{T}(\boldsymbol{Qy}) = \boldsymbol{x}^\mathsf{T}\boldsymbol{y}$. Examples: rotations, reflections, permutations.

**Rotation in 2D** by angle $\theta$ (counterclockwise):

$$\boldsymbol{R}_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$




Columns = images of $\boldsymbol{e}_1, \boldsymbol{e}_2$. $\boldsymbol{R}_\theta^{-1} = \boldsymbol{R}_\theta^\mathsf{T} = \boldsymbol{R}_{-\theta}$, $\det \boldsymbol{R} = 1$, $\boldsymbol{R}_\alpha\boldsymbol{R}_\beta = \boldsymbol{R}_{\alpha+\beta}$.

## Subspaces

**Vector subspace** $S \subseteq \mathbb{R}^n$ is a subspace if $\boldsymbol{0} \in S$ and it is closed under linear combinations:

$$\boldsymbol{x}, \boldsymbol{y} \in S,\ \alpha, \beta \in \mathbb{R} \ \Rightarrow\ \alpha\boldsymbol{x} + \beta\boldsymbol{y} \in S.$$

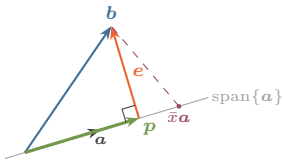
Examples: lines/planes through the origin, $\operatorname{span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\}$, column space $\mathbf{C}(\boldsymbol{A}) = \{\boldsymbol{Ax} \mid \boldsymbol{x} \in \mathbb{R}^n\}$, nullspace $\mathbf{N}(\boldsymbol{A}) = \{\boldsymbol{x} \mid \boldsymbol{Ax} = \boldsymbol{0}, \boldsymbol{x} \in \mathbb{R}^n\}$. A *basis* is an independent spanning set; its size is the *dimension*. $\boldsymbol{x} \perp S$ means $\boldsymbol{x}^\mathsf{T}\boldsymbol{y} = 0$ for all $\boldsymbol{y} \in S$.

## Orthogonal Projection

**Project $b$ onto the line through $a$**

$$\boldsymbol{p} = \frac{\boldsymbol{a}\boldsymbol{a}^\mathsf{T}}{\boldsymbol{a}^\mathsf{T}\boldsymbol{a}}\boldsymbol{b} = \boldsymbol{P}\boldsymbol{b}, \qquad \boldsymbol{P} = \frac{\boldsymbol{a}\boldsymbol{a}^\mathsf{T}}{\boldsymbol{a}^\mathsf{T}\boldsymbol{a}}.$$

Error $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p}$ is **orthogonal** to the line: $\boldsymbol{a}^\mathsf{T}\boldsymbol{e} = 0$ and $\|\boldsymbol{b} - \boldsymbol{p}\|_2^2 = \min_x \|\boldsymbol{b} - \boldsymbol{a}x\|_2^2$.




dashed: $\|\boldsymbol{b} - \bar{x}\boldsymbol{a}\|_2 > \|\boldsymbol{e}\|_2$

**Onto a subspace $\mathbf{C}(A)$**

$$\boldsymbol{p} = \boldsymbol{P}\boldsymbol{b}, \quad \boldsymbol{e} = (\boldsymbol{I} - \boldsymbol{P})\boldsymbol{b}, \quad \boldsymbol{P} = \boldsymbol{A}(\boldsymbol{A}^\mathsf{T}\boldsymbol{A})^{-1}\boldsymbol{A}^\mathsf{T}$$

$\boldsymbol{p} = \boldsymbol{A}\hat{\boldsymbol{x}}$ minimizes the error:

$$\hat{\boldsymbol{x}} = \arg\min_{\boldsymbol{x}} \|\boldsymbol{b} - \boldsymbol{Ax}\|_2^2 = \arg\min_{\boldsymbol{x}} \left(\boldsymbol{x}^\mathsf{T}\boldsymbol{A}^\mathsf{T}\boldsymbol{A}\boldsymbol{x} - 2\boldsymbol{b}^\mathsf{T}\boldsymbol{A}\boldsymbol{x}\right)$$
$$= (\boldsymbol{A}^\mathsf{T}\boldsymbol{A})^{-1}\boldsymbol{A}^\mathsf{T}\boldsymbol{b}$$

Error $\perp$ every column: $\boldsymbol{A}^\mathsf{T}\boldsymbol{e} = \boldsymbol{0}$.

Orthonormal columns ($\boldsymbol{A} = \boldsymbol{Q}$): $\boldsymbol{P} = \boldsymbol{Q}\boldsymbol{Q}^\mathsf{T}$, $\boldsymbol{p} = \sum_i (\boldsymbol{q}_i^\mathsf{T}\boldsymbol{b})\boldsymbol{q}_i$.

**Important properties**

- $(\boldsymbol{A}^\mathsf{T}\boldsymbol{A})^{-1}$ exists iff columns of $\boldsymbol{A}$ are independent

**Properties of a projection matrix $P$**

- $\boldsymbol{P}^2 = \boldsymbol{P}$ (idempotent: projecting twice changes nothing)
- $\boldsymbol{P}^\mathsf{T} = \boldsymbol{P}$ (symmetric); eigenvalues are 0 or 1
- $\boldsymbol{I} - \boldsymbol{P}$ projects onto the orthogonal complement; $\boldsymbol{e} = (\boldsymbol{I} - \boldsymbol{P})\boldsymbol{b}$
- Pythagoras: $\|\boldsymbol{b}\|_2^2 = \|\boldsymbol{p}\|_2^2 + \|\boldsymbol{e}\|_2^2$, so $\|\boldsymbol{Pb}\|_2 \leq \|\boldsymbol{b}\|_2$
- $\boldsymbol{p}$ is the unique closest point of the subspace to $\boldsymbol{b}$

# Math for ML Cheat Sheet: Eigenvalues, Calculus & Optimization

References: Strang ILA • Deisenroth et al. MML

## Eigenvalues & Eigenvectors

For square $\boldsymbol{A} \in \mathbb{R}^{n\times n}$, a nonzero $\boldsymbol{x}$ is an **eigenvector** with **eigenvalue** $\lambda$ if

$$\boldsymbol{A}\boldsymbol{x} = \lambda\boldsymbol{x} \qquad (\boldsymbol{x} \neq \boldsymbol{0}).$$

$\boldsymbol{A}$ only stretches $\boldsymbol{x}$ by $\lambda$, never changes its direction (up to sign).

- eigenvalues solve $\det(\boldsymbol{A} - \lambda\boldsymbol{I}) = 0$
- $\sum_i \lambda_i = \operatorname{tr}\boldsymbol{A}$, $\prod_i \lambda_i = \det\boldsymbol{A}$
- $\boldsymbol{A}^k\boldsymbol{x} = \lambda^k\boldsymbol{x}$; $\boldsymbol{A}^{-1}\boldsymbol{x} = \lambda^{-1}\boldsymbol{x}$

**Diagonalization** If $\boldsymbol{A}$ has $n$ *independent* eigenvectors (columns of $\boldsymbol{X}$), eigenvalues on the diagonal of $\boldsymbol{\Lambda}$:

$$\boldsymbol{A} = \boldsymbol{X}\boldsymbol{\Lambda}\boldsymbol{X}^{-1}, \qquad \boldsymbol{A}^k = \boldsymbol{X}\boldsymbol{\Lambda}^k\boldsymbol{X}^{-1}.$$




**Not always possible!** E.g. $\left[\begin{smallmatrix}0&1\\0&0\end{smallmatrix}\right]$ has $\lambda = 0$ twice but only one eigenvector direction. Guaranteed when the $n$ eigenvalues are distinct and real, or $\boldsymbol{A}$ is symmetric. Eigenvalues may be complex (e.g. rotations).

**Symmetric matrices** (spectral theorem) If $\boldsymbol{A} = \boldsymbol{A}^\mathsf{T}$: all $\lambda_i$ real, eigenvectors can be chosen orthonormal, so $\boldsymbol{X} = \boldsymbol{Q}$ is orthogonal:

$$\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^\mathsf{T} = \sum_{i=1}^{n} \lambda_i\, \boldsymbol{q}_i\boldsymbol{q}_i^\mathsf{T}.$$




## Definiteness & Quadratic Forms

**Quadratic form** For symmetric $\boldsymbol{A} \in \mathbb{R}^{n\times n}$:

$$\boldsymbol{x}^\mathsf{T}\boldsymbol{A}\boldsymbol{x} = \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}x_ix_j$$

$$\begin{bmatrix}x_1\\x_2\end{bmatrix}^\mathsf{T}\begin{bmatrix}a & b\\ b & c\end{bmatrix}\begin{bmatrix}x_1\\x_2\end{bmatrix} = ax_1^2 + 2b\,x_1x_2 + c\,x_2^2.$$

Non-symmetric $\boldsymbol{A}$: same form as $\frac{1}{2}(\boldsymbol{A} + \boldsymbol{A}^\mathsf{T})$. With $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^\mathsf{T}$ it is a **sum of squares** along the eigenvectors:

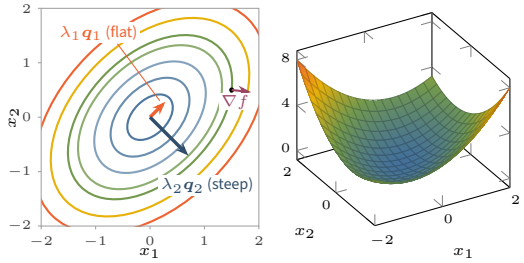
$$\boldsymbol{x}^\mathsf{T}\boldsymbol{A}\boldsymbol{x} = \sum_{i=1}^{n} \lambda_i\,(\boldsymbol{q}_i^\mathsf{T}\boldsymbol{x})^2.$$

**Positive (semi-)definite** Symmetric $\boldsymbol{A}$ is

- **PD** ($\boldsymbol{A} \succ 0$): $\boldsymbol{x}^\mathsf{T}\boldsymbol{A}\boldsymbol{x} > 0$ for all $\boldsymbol{x} \neq \boldsymbol{0} \iff$ all $\lambda_i > 0$
- **PSD** ($\boldsymbol{A} \succeq 0$): $\boldsymbol{x}^\mathsf{T}\boldsymbol{A}\boldsymbol{x} \geq 0$ for all $\boldsymbol{x} \iff$ all $\lambda_i \geq 0$

$\boldsymbol{A}^\mathsf{T}\boldsymbol{A}$ is always PSD (PD if the columns of $\boldsymbol{A}$ are independent), since $\boldsymbol{x}^\mathsf{T}\boldsymbol{A}^\mathsf{T}\boldsymbol{A}\boldsymbol{x} = \|\boldsymbol{A}\boldsymbol{x}\|_2^2$. PD $\Rightarrow$ invertible.

**Picture of a quadratic** $f(\boldsymbol{x}) = \boldsymbol{x}^\mathsf{T}\boldsymbol{A}\boldsymbol{x}$ with $\boldsymbol{A} = \left[\begin{smallmatrix}0.7 & -0.3\\ -0.3 & 0.7\end{smallmatrix}\right]$, $\lambda_1 = 0.4$, $\boldsymbol{q}_1 = \frac{1}{\sqrt{2}}\left[\begin{smallmatrix}1\\1\end{smallmatrix}\right]$, $\lambda_2 = 1$, $\boldsymbol{q}_2 = \frac{1}{\sqrt{2}}\left[\begin{smallmatrix}1\\-1\end{smallmatrix}\right]$ (cf. `code/julia/10-quadratic.jl`)




Level sets $\boldsymbol{x}^\mathsf{T}\boldsymbol{A}\boldsymbol{x} = c$ are ellipses with axes along $\boldsymbol{q}_i$ and half-lengths $\sqrt{c/\lambda_i}$: small $\lambda$ = long, flat direction. Gradient is $\perp$ to level sets. PD: bowl; PSD: valley; indefinite: saddle.

## Derivatives

**Partial derivative** of $f : \mathbb{R}^n \to \mathbb{R}$ w.r.t. $x_i$ (others fixed):

$$\frac{\partial f}{\partial x_i}(\boldsymbol{x}) = \lim_{h\to 0}\frac{f(\boldsymbol{x} + h\boldsymbol{e}_i) - f(\boldsymbol{x})}{h}$$

**Gradient** column vector of partials

$$\nabla f(\boldsymbol{x}) = \left[\frac{\partial f}{\partial x_1}(\boldsymbol{x}) \quad \cdots \quad \frac{\partial f}{\partial x_n}(\boldsymbol{x})\right]^\mathsf{T} \in \mathbb{R}^n$$

Points in the direction of steepest ascent; directional derivative along $\boldsymbol{d}$ is $\nabla f(\boldsymbol{x})^\mathsf{T}\boldsymbol{d}$. MML writes the row vector $\frac{df}{d\boldsymbol{x}} = \nabla f(\boldsymbol{x})^\mathsf{T}$.

**Hessian** $\nabla^2 f(\boldsymbol{x}) \in \mathbb{R}^{n\times n}$, $[\nabla^2 f(\boldsymbol{x})]_{ij} = \dfrac{\partial^2 f}{\partial x_i \partial x_j}$, symmetric when $f$ is twice continuously differentiable.

**Jacobian** of $\boldsymbol{g} : \mathbb{R}^n \to \mathbb{R}^m$: $\boldsymbol{J}_{\boldsymbol{g}}(\boldsymbol{x}) \in \mathbb{R}^{m\times n}$, $[\boldsymbol{J}_{\boldsymbol{g}}]_{ij} = \partial g_i/\partial x_j$. Row $i$ is the gradient of the component $g_i$, transposed.




$\nabla f \in \mathbb{R}^n$ | $\nabla^2 f \in \mathbb{R}^{n\times n}$ symmetric | $\boldsymbol{J}_{\boldsymbol{g}} \in \mathbb{R}^{m\times n}$

**Example** $\boldsymbol{g}(\boldsymbol{x}) = \left[\begin{smallmatrix}x_1^2x_2\\ x_1 + 3x_2\end{smallmatrix}\right]$, $\boldsymbol{g} : \mathbb{R}^2 \to \mathbb{R}^2$:

$$\nabla g_1 = \begin{bmatrix}2x_1x_2\\ x_1^2\end{bmatrix},\ \nabla g_2 = \begin{bmatrix}1\\3\end{bmatrix} \Rightarrow \boldsymbol{J}_{\boldsymbol{g}} = \begin{bmatrix}\nabla g_1^\mathsf{T}\\ \nabla g_2^\mathsf{T}\end{bmatrix} = \begin{bmatrix}2x_1x_2 & x_1^2\\ 1 & 3\end{bmatrix}$$

The Hessian is the Jacobian of the gradient: $\nabla^2 g_1 = \boldsymbol{J}_{\nabla g_1} = \left[\begin{smallmatrix}2x_2 & 2x_1\\ 2x_1 & 0\end{smallmatrix}\right]$.

**Linear & quadratic functions** $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^n$, $\boldsymbol{A} \in \mathbb{R}^{n\times n}$, $\boldsymbol{B} \in \mathbb{R}^{m\times n}$, $\boldsymbol{y} \in \mathbb{R}^m$

| $f(\boldsymbol{x})$ | $\nabla f(\boldsymbol{x})$ | $\nabla^2 f(\boldsymbol{x})$ |
|---|---|---|
| $\boldsymbol{a}^\mathsf{T}\boldsymbol{x} + c$ | $\boldsymbol{a}$ | $\boldsymbol{0}$ |
| $\boldsymbol{x}^\mathsf{T}\boldsymbol{A}\boldsymbol{x}$ | $(\boldsymbol{A} + \boldsymbol{A}^\mathsf{T})\boldsymbol{x}$ | $\boldsymbol{A} + \boldsymbol{A}^\mathsf{T}$ |
| $\boldsymbol{x}^\mathsf{T}\boldsymbol{A}\boldsymbol{x}$, $\boldsymbol{A}$ sym. | $2\boldsymbol{A}\boldsymbol{x}$ | $2\boldsymbol{A}$ |
| $\frac{1}{2}\boldsymbol{x}^\mathsf{T}\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}^\mathsf{T}\boldsymbol{x}$, sym. | $\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}$ | $\boldsymbol{A}$ |
| $\|\boldsymbol{B}\boldsymbol{x} - \boldsymbol{y}\|_2^2$ | $2\boldsymbol{B}^\mathsf{T}(\boldsymbol{B}\boldsymbol{x} - \boldsymbol{y})$ | $2\boldsymbol{B}^\mathsf{T}\boldsymbol{B}$ |

Vector-valued: $\boldsymbol{g}(\boldsymbol{x}) = \boldsymbol{B}\boldsymbol{x}$ has Jacobian $\boldsymbol{J}_{\boldsymbol{g}} = \boldsymbol{B}$.

**Chain rule, 1-d** $h(x) = f(g(x))$:

$$h'(x) = f'(g(x))\,g'(x)$$

**Chain rule, multi-d** $h(\boldsymbol{x}) = f(\boldsymbol{g}(\boldsymbol{x}))$, $\boldsymbol{g} : \mathbb{R}^n \to \mathbb{R}^m$, $f : \mathbb{R}^m \to \mathbb{R}$:

$$\nabla h(\boldsymbol{x}) = \boldsymbol{J}_{\boldsymbol{g}}(\boldsymbol{x})^\mathsf{T}\,\nabla f(\boldsymbol{g}(\boldsymbol{x}))$$

MML (row form): $\frac{dh}{d\boldsymbol{x}} = \frac{df}{d\boldsymbol{g}}\frac{d\boldsymbol{g}}{d\boldsymbol{x}}$, sizes $(1\times n) = (1\times m)(m\times n)$. Useful cases: $\nabla_{\boldsymbol{x}} f(\boldsymbol{B}\boldsymbol{x} + \boldsymbol{y}) = \boldsymbol{B}^\mathsf{T}\nabla f(\boldsymbol{B}\boldsymbol{x} + \boldsymbol{y})$; along a line, $\frac{d}{dt}f(\boldsymbol{x} + t\boldsymbol{d}) = \nabla f(\boldsymbol{x} + t\boldsymbol{d})^\mathsf{T}\boldsymbol{d}$.

## Taylor Approximation

$g(\boldsymbol{d}) = o(\|\boldsymbol{d}\|_2^k)$ means $g(\boldsymbol{d})/\|\boldsymbol{d}\|_2^k \to 0$ as $\boldsymbol{d} \to \boldsymbol{0}$ (smaller than $\|\boldsymbol{d}\|_2^k$).

**Linear (1st order)**

$$f(\boldsymbol{x} + \boldsymbol{d}) = f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^\mathsf{T}\boldsymbol{d} + o(\|\boldsymbol{d}\|_2)$$

**Quadratic (2nd order)**

$$f(\boldsymbol{x} + \boldsymbol{d}) = f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^\mathsf{T}\boldsymbol{d} + \tfrac{1}{2}\boldsymbol{d}^\mathsf{T}\nabla^2 f(\boldsymbol{x})\,\boldsymbol{d} + o(\|\boldsymbol{d}\|_2^2)$$

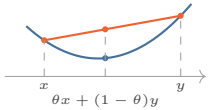
1-d: $f(x + d) = f(x) + f'(x)\,d + \frac{1}{2}f''(x)\,d^2 + o(d^2)$.

## Convexity

**Convex combination** of $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_k$:

$\sum_i \theta_i\boldsymbol{x}_i$, $\theta_i \geq 0$, $\sum_i \theta_i = 1$.

For two points: $\theta\boldsymbol{x} + (1-\theta)\boldsymbol{y}$, $\theta \in [0,1]$ (the segment). A set is convex if it contains all convex combinations of its points.




chord lies above the graph

**Convex function** $f$ (on a convex domain) if for all $\boldsymbol{x}, \boldsymbol{y}, \theta \in [0,1]$:

$$f(\theta\boldsymbol{x} + (1-\theta)\boldsymbol{y}) \leq \theta f(\boldsymbol{x}) + (1-\theta)f(\boldsymbol{y}).$$

1-d is the same with scalars $x, y$. *Strictly* convex: $<$ for $\boldsymbol{x} \neq \boldsymbol{y}, \theta \in (0,1)$.

**Rules for checking convexity**

- 1-d: $f''(x) \geq 0$ for all $x \iff f$ convex (twice differentiable)
- multi-d: $\nabla^2 f(\boldsymbol{x}) \succeq 0$ (PSD) for all $\boldsymbol{x} \iff f$ convex; $\nabla^2 f \succ 0 \Rightarrow$ strictly convex
- 1st order: $f(\boldsymbol{y}) \geq f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^\mathsf{T}(\boldsymbol{y} - \boldsymbol{x})$ (tangent below graph)
- preserved by: nonnegative sums $\alpha f + \beta g$, composition with affine map $f(\boldsymbol{B}\boldsymbol{x} + \boldsymbol{y})$, pointwise $\max\{f, g\}$
- e.g. $\boldsymbol{x}^\mathsf{T}\boldsymbol{A}\boldsymbol{x}$ convex $\iff \boldsymbol{A} \succeq 0$; every norm is convex

## Optimality & Gradient Descent

**Conditions for a local minimum** $x^\star$ ($f$ twice differentiable)

- **necessary:** $\nabla f(\boldsymbol{x}^\star) = \boldsymbol{0}$ and $\nabla^2 f(\boldsymbol{x}^\star) \succeq 0$
- **sufficient:** $\nabla f(\boldsymbol{x}^\star) = \boldsymbol{0}$ and $\nabla^2 f(\boldsymbol{x}^\star) \succ 0$ (strict local min)
- **convex** $f$: $\nabla f(\boldsymbol{x}^\star) = \boldsymbol{0} \iff \boldsymbol{x}^\star$ is a *global* minimum

1-d: $f'(x^\star) = 0$, and $f''(x^\star) \geq 0$ (necessary) / $> 0$ (sufficient). Ex.: $\min \frac{1}{2}\boldsymbol{x}^\mathsf{T}\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}^\mathsf{T}\boldsymbol{x}, \boldsymbol{A} \succ 0 \Rightarrow \boldsymbol{x}^\star = \boldsymbol{A}^{-1}\boldsymbol{b}$.

**Gradient descent** from $\boldsymbol{x}_0$ with step size (learning rate) $\alpha > 0$:

$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k - \alpha\,\nabla f(\boldsymbol{x}_k)$$

Moves against the gradient (steepest descent). Too large $\alpha$ diverges; for $\frac{1}{2}\boldsymbol{x}^\mathsf{T}\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}^\mathsf{T}\boldsymbol{x}$ it converges iff $0 < \alpha < 2/\lambda_{\max}(\boldsymbol{A})$.

